

High Resolution Technologies

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Acknowledgement Of Country

I'd like to acknowledge the Kaurna people as the traditional owners and custodians of the land we know today as the Adelaide Plains, where I live & work.

I also acknowledge the deep feelings of attachment and relationship of the Kaurna people to their place.

I pay my respects to the cultural authority of Aboriginal and Torres Strait Islander peoples from other areas of Australia, and pay my respects to Elders past, present and emerging, and acknowledge any Aboriginal Australians who may be with us today

Introduction

High Resolution Technologies

- Microarray and early RNA-Seq analysis used 'bulk' tissues
- Very hard to obtain pure cell-types in most samples
 - T cells sorted by FACS still heterogeneous
 - Cancer biopsies are heterogeneous
 - Immortalised cell-lines (HeLa, HEK293 etc.) differentiate into multiple cell types
- Single-Cell RNA-Seq (scRNA-Seq) $\Rightarrow$ RNA can be associated with "cell of origin"

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- Spatial Transcriptomics $\Rightarrow$ Identify cell-types on a slide by location
 - Doesn't always utilise sequencing

High Resolution Technologies

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 - And where are the transcripts within a cell?
- Which cells are communicating with which other cells?

Single Cell RNA-Seq

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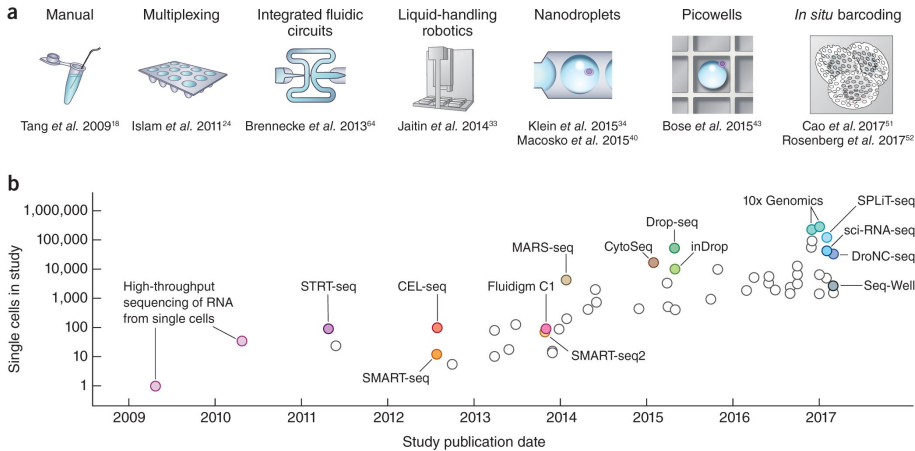


Figure 1: Figure from Svensson, Vento-Tormo, and Teichmann (2018)

Single Cell RNA-Seq

Four basic scRNA technologies

- ① Droplet-based (e.g. 10x Genomics, Drop-Seq):
 - Cells captured in micro-droplets → barcoded → lysed
- ② Plate-based (e.g. SMART-Seq2)
 - Individual cells captured in *microwells*
- ③ Gel/Emulsification-based
 - Similar to droplet but performed in solution
- ④ Pooling & Splitting (e.g. SPLIT-Seq)
 - Multiple wells with distinct barcodes
 - Cells pooled → split → next barcode

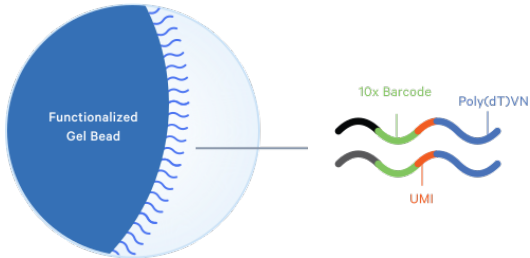
Single Cell RNA-Seq

Table 1: Data sourced from Haque et al. (2017)

Protocol	C1 (SMART-Seq)	SMART-Seq2	10X Chromium	SPLIT-Seq
<i>Platform</i>	Microfluidics	Plate-based	Droplet	Plate-based
<i>Transcript</i>	Full-length	Full-length	3'-end	3'-end
<i>Cells</i>	$10^2 - 10^3$	$10^2 - 10^3$	$10^3 - 10^4$	$10^3 - 10^5$
<i>Reads/Cell</i>	10^6	10^6	$10^4 - 10^5$	10^4

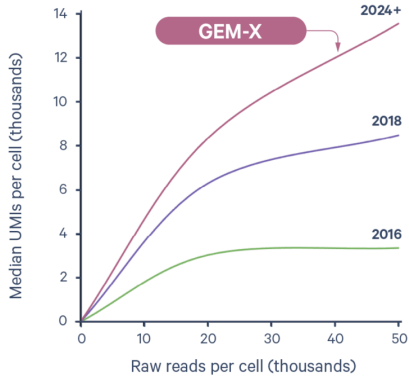
Tradeoff is between *cell numbers*, *transcripts per cell*, *full length vs 3'*, *cost* etc

Droplet-Based scRNA



- GEM: Gel bead in EMulsion
- Primers can capture 3', 5' or targeted sequences
- Figures from www.10xgenomics.com

Droplet-Based scRNA



- UMIs represent individual RNA molecules
- Now capturing ~14,000 transcripts/cell

Figure 2: www.10xgenomics.com

Data Processing

- Most pre-processing for 10X data is performed using CellRanger
- Handles demultiplexing, alignment (STAR) and quantification (using UMIs)
 - Full-length transcript methods can utilise kallisto/salmon
- We end up with a feature-barcode matrix
 - A barcode represents an individual cell (or a set of reactions)
 - A feature is commonly thought of as a gene in scRNA-Seq
- Similar to counts from bulk RNA-Seq but with many more columns (cells)
 - Also many more zero counts for genes
 - Multiple cells from multiple biological replicates

Filtering Stages

- The aim is to keep the *high quality cells* and discard the dubious cells, such as:
 - ① Low/High read numbers (library sizes)
 - ② Low feature/gene numbers
 - ③ High proportions of mitochondrial RNA $\implies$ cells broken prior to lysis
 - ④ Doublets (i.e. two cells instead of a single cell)
- Also filter for genes considered as detectable (Average Counts > 1)
 - Some zero counts are from unexpressed genes
 - Others are “dropouts” $\rightarrow$ RNA capture efficiency between 10-40%

Normalisation

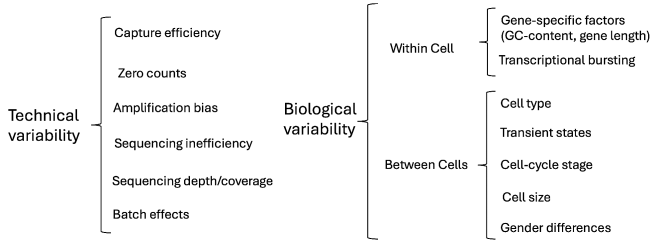


Figure 3: Image from Cuevas-Diaz Duran, Wei, and Wu (2024)

- Bulk-RNA normalisation methods no longer applicable
 - Does each cell (or cell-type) have the same starting amount of RNA?
- Normalisation strategies may be platform dependent
 - Gene length impacts full-length methods but not droplet-based (3' or 5')
- Should missing values be imputed?

Identifying Cell Types

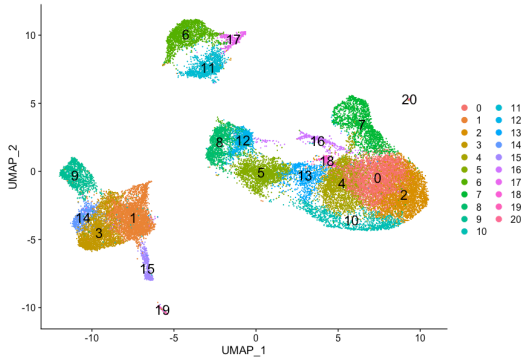


Figure 4: Image courtesy of the Harvard Chan Bioinformatics Core

- A key aim is to identify different cell types in our sample
- How many cell types are there really?
- How similar are cell types?
- What genes drive the differences → Differential Expression
- ① Dimensional Reduction

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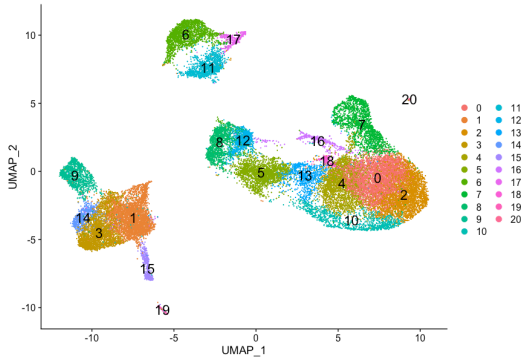


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 - ② Clustering Cells

Principal Component Analysis

- scRNA (+ bulk RNA) are high dimensional datasets
 - e.g. 1,000 cells $\times$ 15,000 genes
- How do we see how similar cells are to each other?
 - Is there a way to summarise distances between cells?
 - Can we visualise in 2-dimensions?
- Principal Component Analysis (PCA) identifies direction of maximum variance

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- Data is rotated around PC1, then PC2 etc
- Comparing PC1 vs PC2, PC2 vs PC3 etc allows 2D (or 3D) visualisation

Principal Component Analysis

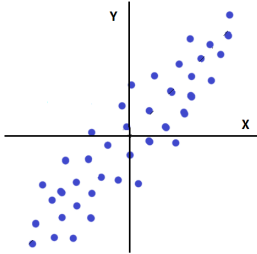


Figure 5: Some 2D data points

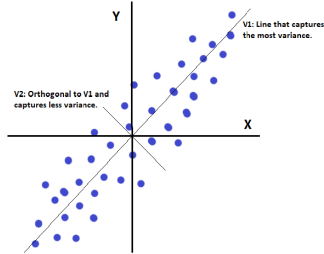


Figure 6: Find the direction of max variance

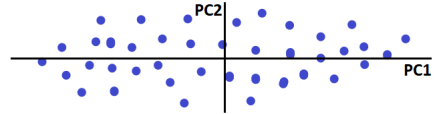


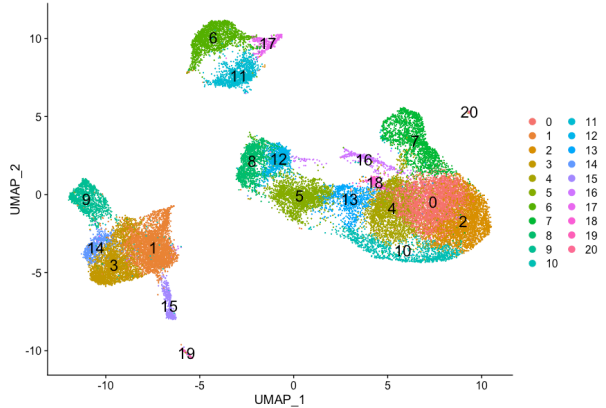
Figure 7: Each point now has a 'rotated' value

- The main directions of variability are hopefully related to the biology
- Rotations along PC1 vs PC2 should show separation by biology of interest

Figures taken from <https://statisticsbyjim.com/basics/principal-component-analysis/>

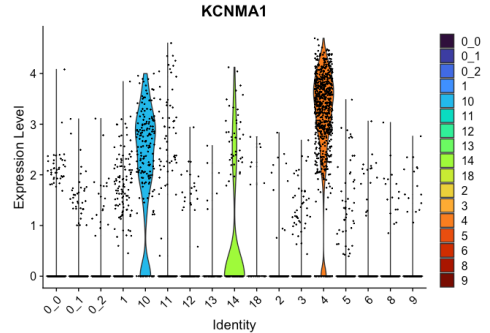
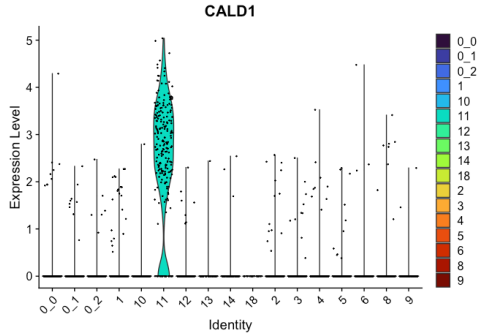
Clustering of Cells

- To form clusters $\Rightarrow$ use top principal components
- Often use k-nearest-neighbour (kNN) graph methods to form clusters
- Methods like UMAP highlight clusters during visualisation



- Huge number of possible parameters $\Rightarrow$ subjective approach to data

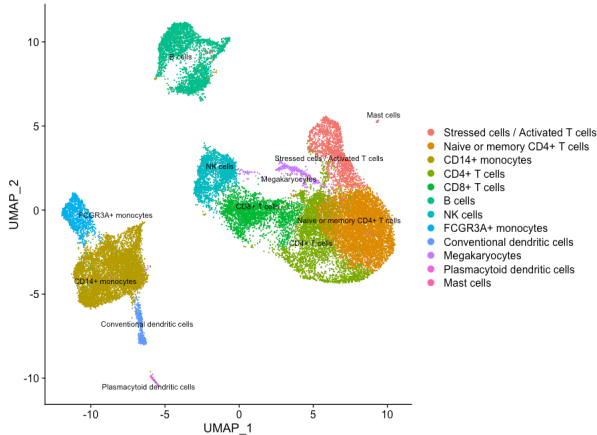
Marker Detection



- Find genes unique to a given cluster or shared across multiple clusters

Figures taken from <https://ucdavis-bioinformatics-training.github.io/2023-June-Single-Cell-RNA-Seq-Analysis>

Cluster Annotation



- Use our expert knowledge of known marker genes to annotate clusters
- CITE-Seq uses cell-surface antibodies to aid this step
- Can drill deeply into cell-type differences & communication
 - Complementary receptor-ligand expression

Figure 8: Image courtesy of the Harvard Chan Bioinformatics Core

Spatial Transcriptomics

Spatial Transcriptomics

- Even single cell RNA dissociates cells from position with tissues
- Cells function within highly structured tissue environments
 - Cells are likely to have different communication dynamics based on location
- *Spatial transcriptomics* attempts to resolve this limitation

Early Spatial Transcriptomics

- First appeared using barcoded primers with fixed location (Ståhl et al. 2016)
 - $\sim 100 \mu\text{m}$ resolution
- Tissue affixed to prepared slide:
 - Stained & Visualised
 - Made Permeable
 - RT-PCR $\Rightarrow$ cDNA
 - Tissue removed enzymatically
 - Barcoded cDNA $\Rightarrow$ sequencing

Approaching Single Cell Resolution

- 10X Visium now down to $100\ \mu\text{m}$ resolution
- *Slide-seq* (Stickels et al. 2021): Barcoded beads $\rightarrow \sim 10\ \mu\text{m}$ resolution
 - Approaching single cell resolution
 - Deconvolution of beads single cell
- Stereo-Seq (A. Chen et al. 2022): etched grid on slide $\Rightarrow$ resolution of $\sim 200\text{nm}$
- Seq-Scope (Cho et al. 2021): Tissue placed directly on modified Illumina flow-cell $\Rightarrow$ resolution $\sim 500\text{nm}$

Beyond Sequencing

- Barcoded probes via Fluorescence In Situ Hybridisation enable a sequencing-free perspective
- Staining technologies used to identify cell boundaries, nuclei etc
- MERFISH: multiplexed error-robust FISH (K. H. Chen et al. 2015)
 - Sequence specific barcodes able to detect ~10,000 transcripts
 - Directly count specific transcripts
 - Location within cell also identified
- Extremely rapid technological development (also \$\$\$)
- Methods often lag technology somewhat

Examples of 10X Xenium Spatial Transcriptomics

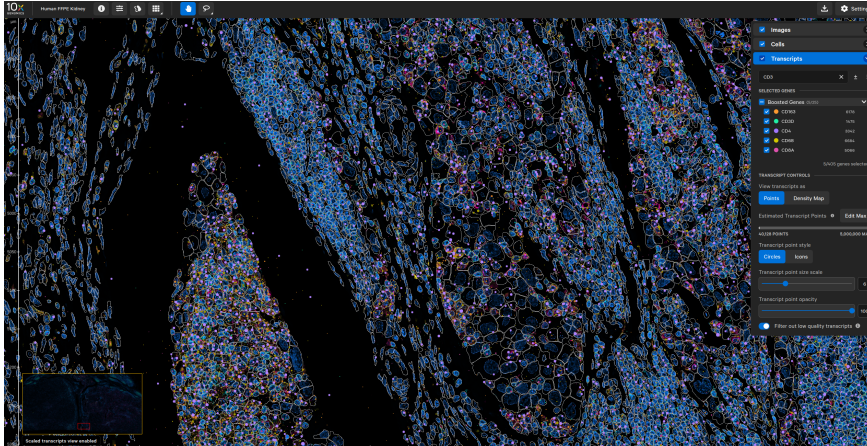
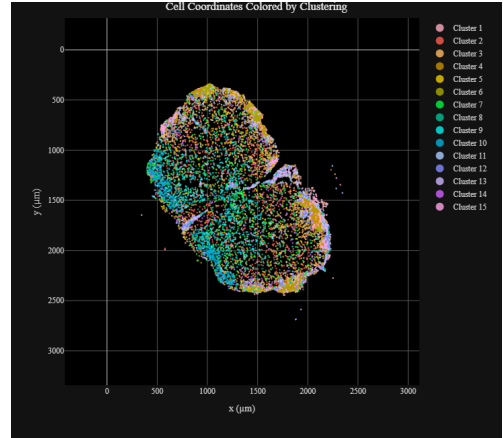
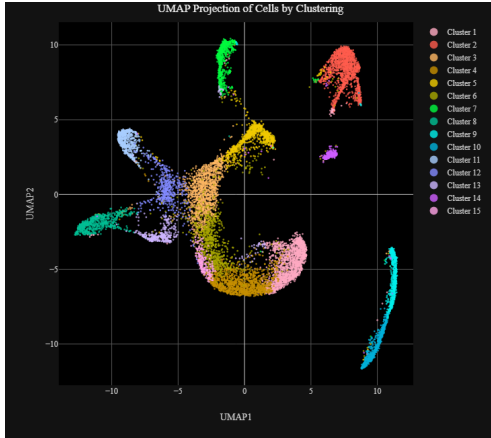


Figure 9: Taken from <https://xenium.10xgenomics.com>. Highlighting CD4 in human kidney

Examples of 10X Xenium Spatial Transcriptomics



Images courtesy of Megan Monaghan. PhD Candidate, Comerford Lab. Adelaide University

Additional Reading

- StatQuest: Principal Component Analysis (PCA)
- StatQuest: K-nearest neighbors, Clearly Explained
- StatQuest: UMAP Dimension Reduction, Main Ideas!!!

References

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